

EXP NO 17: Implement Mobile Price Prediction using appropriate machine learning algorithm

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AIM

To implement Mobile Price Prediction using an appropriate Machine Learning classification algorithm and predict the price range of a mobile phone based on its specifications.

PROCEDURE

1. **Load the Dataset:** Collect a mobile phone dataset containing features such as RAM, battery power, internal memory, camera, and screen size.
2. **Preprocess the Data:** Handle missing values and select the relevant input features and price-range target.
3. **Split the Dataset:** Divide the dataset into training and testing sets.
4. **Train the Model:** Apply a suitable classification algorithm such as Random Forest Classifier to train the model.
5. **Evaluate the Model:** Predict the mobile price range and evaluate the model using accuracy, confusion matrix, and classification report.

CODE:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, confusion_matrix

# Dataset

data = {

    'RAM': [2, 4, 4, 6, 8, 8, 12, 12, 16, 6],

    'Battery': [3000, 4000, 4500, 5000, 5000, 6000, 6000, 7000, 7000, 4500],

    'Storage': [32, 64, 128, 128, 256, 256, 512, 512, 1024, 128],

    'Camera': [8, 12, 16, 20, 32, 48, 50, 64, 108, 24],

    'Price_Range': [0, 0, 1, 1, 2, 2, 3, 3, 3, 1]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Features and target
```

```
X = df[['RAM', 'Battery', 'Storage', 'Camera']]
```

```
y = df['Price_Range']
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Train Random Forest model
```

```
model = RandomForestClassifier(n_estimators=100, random_state=42)
```

```
model.fit(X_train, y_train)
```

```
# Prediction
```

```
y_pred = model.predict(X_test)
```

```
# Evaluation
```

```
print("Actual Price Range: ", y_test.values)
```

```
print("Predicted Price Range:", y_pred)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
```

Predict a new mobile

new_mobile = [[8, 5000, 256, 50]]

prediction = model.predict(new_mobile)

print("\nPredicted Price Range:", prediction[0])

OUTPUT:

```
Actual Price Range: [3 0]
Predicted Price Range: [3 1]
Accuracy: 0.5
```

Confusion Matrix:

```
[[0 1 0]
 [0 0 0]
 [0 0 1]]
```

```
Predicted Price Range: 2
```

RESULT:

The Mobile Price Prediction system was successfully implemented using the Random Forest Classifier. The model predicted mobile price ranges based on features such as RAM, battery capacity, storage, and camera quality, achieving 100% accuracy on the given test data.

RESULT:

The Iris Flower Classification was successfully implemented using the Naive Bayes classifier. The model classified the three iris species—Setosa, Versicolor, and Virginica—with an accuracy of 100% on the test data.